Developing an in silico transmission model of Vaccinopolis, a clinical research facility, to support the design of human infection trials

Master's Thesis Defence

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Outline

1 Background & Objective

2 Methodology

3 Results

4 Calibration & Sensitivity Analysis

5 Discussion & Conclusion

6 Limitations

Vaccinopolis: Background





Research context

Vaccinopolis is an innovative vaccine research center located in Antwerp.

- An ambulatory unit for vaccine trials.
- A quarantine unit for human infection trials (maximum 30 rooms).

Problems

- Experimental trials are expensive and risky.
- Understanding transmission risk during human infectious trials.
- Limited empirical data. Lack of models that combine human behavior with biological data.

Vaccinopolis: Study setting and Objective

Research question: *How can infectious disease transmission within the Vaccinopolis center be simulated using an individual-based model, integrating interpersonal contact patterns, individual viral progression, and individual-level heterogeneity in susceptibility?*

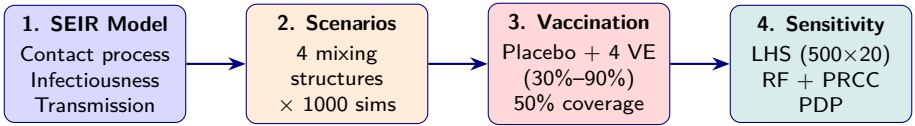
Study setting

- $n = 28$: **24 volunteers + 4 staff**
- Staff: clinical monitoring, sample collection, meal service, and daily operations.
- Volunteers: reside on quarantine floor, sharing common living areas.
- Different daily schedules $\Rightarrow$ different contact patterns.

Objectives

1. Map interpersonal contact patterns
2. Model infectiousness dynamics $\nu(t)$
3. Simulate person-to-person transmission
4. Evaluate outbreak outcomes under different containment strategies

Methodology Pipeline



Outcomes

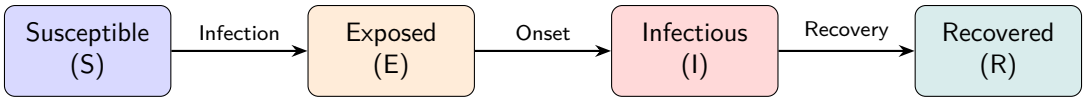
- Daily infection incidence: $I_t = \sum_{i=1}^{N} 1(t_i \in [t, t+1))$
- Attack rate (AR) distributions $AR_{\text{Total}} = \frac{N_{\text{infected}}}{N}$
- Vaccination impact on AR: $z_{sus,i}^{vacc} = z_{sus,i} \cdot (1 - \varepsilon_{sus})$

Calibration & Sensitivity analysis

Time-block calibration check
7 parameters varied via LHS
Outcomes: $logit(AR_{total}, AR_{vol}, AR_{staff})$
Sensitivity analysis methods:

- Random Forest variable importance (%IncMSE)
- Partial rank correlation coefficients (PRCC)
- Partial dependence plots (PDP)

SEIR Model Framework



SEIR model state transitions

Key components

Contact process: Poisson, time-varying $\lambda_i(t)$

Infectiousness: $\nu(t) \sim$ Gamma

Transmission:

$p_{ij}(t) = Q \cdot z_{\text{inf},i} \cdot \nu(t - t_i^{\text{onset}}) \cdot z_{\text{sus},j}$

Calibration: $Q = \frac{R_0}{\lambda \times M_{total}}$

Individual heterogeneity

$z_{\text{inf}} \sim \text{Gamma}(k, 1/k)$: infectiousness

$z_{\text{sus}} \sim \text{Gamma}(k_{\text{sus}}, 1/k_{\text{sus}})$: susceptibility

$\lambda_{0,i} \sim \text{Gamma}(k_{\text{ct}}, \text{ct.rates}/k_{\text{ct}})$: baseline contact rates

Low k $\Rightarrow$ superspreading dynamics

Time-Varying Contact Rates

Contact multipliers of Staff and Volunteers

Time block	Hours	Period	Staff	Vol.
Morning	5h	06:00–11:00	1.5	1.2
Lunch	2h	11:00–13:00	1.0	2.0
Afternoon	5h	13:00–18:00	1.5	1.0
Dinner	2h	18:00–20:00	1.0	1.8
Evening	2h	20:00–22:00	0.8	1.5
Night	8h	22:00–06:00	0.6	0.5

- Transmission rate at time t:

$$\lambda_i(t) = \lambda_{0,i} \times M_{\text{role}}(\text{block}(t))$$

- Average multiplier for each role over a 24-hour period:

$$\bar{M}_{role} = \frac{\sum_{i=1}^{6}(M_{i,role} \times t_i)}{24}$$

- Overall population adjustment factor:

$$\bar{M}_{total} = \frac{(\bar{M}_{staff} \times N_{staff}) + (\bar{M}_{vol} \times N_{vol})}{N_{total}} \approx 1.065$$

Staff: higher during working hours
Volunteers: higher during meals & evening

Multipliers are assumptions; calibration preserves R_0

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Four Mixing Scenarios

- **Scenario 1**: Free mixing
  $P(j|i) = 1/27$
- **Scenario 2**: Staff-bridge
  2 rooms + shared staff
- **Scenario 3**: Strict cohorting
  2 independent cohorts
- **Scenario 4**: Common staff room
  Scenario 3 + 10% inter-staff

m_{hg}: probability that a contact made by an individual in group h (row) takes place with an individual in group g (column). Rows sum to 1.

Contact probability matrix for each scenario

Scenario 2

	Vol A	Vol B	Staff
Vol A	0.90	0.00	0.10
Vol B	0.00	0.90	0.10
Staff	0.40	0.40	0.20

Scenario 3

	Vol A	Vol B	Staff A	Staff B
Vol A	0.90	0.00	0.10	0.00
Vol B	0.00	0.90	0.00	0.10
Staff A	0.50	0.00	0.50	0.00
Staff B	0.00	0.50	0.00	0.50

Scenario 4

	Vol A	Vol B	Staff A	Staff B
Vol A	0.90	0.00	0.10	0.00
Vol B	0.00	0.90	0.00	0.10
Staff A	0.45	0.00	0.45	0.10
Staff B	0.00	0.45	0.10	0.45

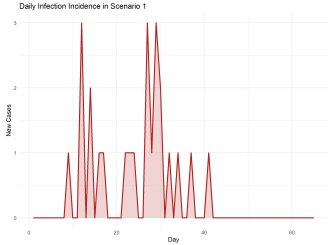
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Model Parameters

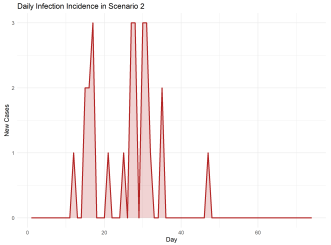
Model parameters and sensitivity analysis range

Parameter	Mean (SD)	Distribution	SA Range
Population size n	28	NA	–
Basic repro. number R_0	1.5	NA	$[1.5, 4.0]$
Latent period *mean*$_e$	5.2 (2.8) days	Weibull	$[3.0, 7.0]$
Recovery period	18 days	Exponential	–
Daily contact rate *ct.rates*	3 (2.12) days	Gamma	$[1.0, 6.0]$
Infectiousness *mean*$_i$	10 (3.8) days	Gamma	$[5.0, 12.0]$
Infectivity heterogeneity k	0.2 (2.236)	Gamma $(k, 1/k)$	$[0.1, 1.0]$
Individual infectivity z_{inf}	1 (1.414)	via k	–
Susceptibility heterogeneity k_{sus}	1 (1)	Gamma $(k_{\text{sus}}, 1/k_{\text{sus}})$	$[0.5, 1.5]$
Individual susceptibility z_{sus}	1 (1)	via k_{sus}	–
Contact heterogeneity k_{ct}	1 (1)	Gamma $(k_{\text{ct}}, 1/k_{\text{ct}})$	$[0.5, 1.5]$

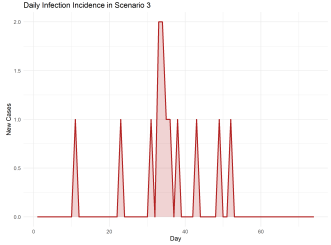
Results: Daily Infection Incidence



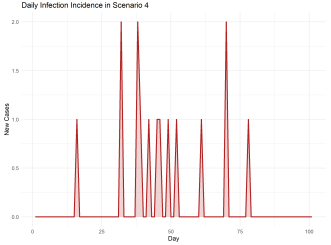
(a) Scenario 1: Free mixing



(b) Scenario 2: Staff-bridge

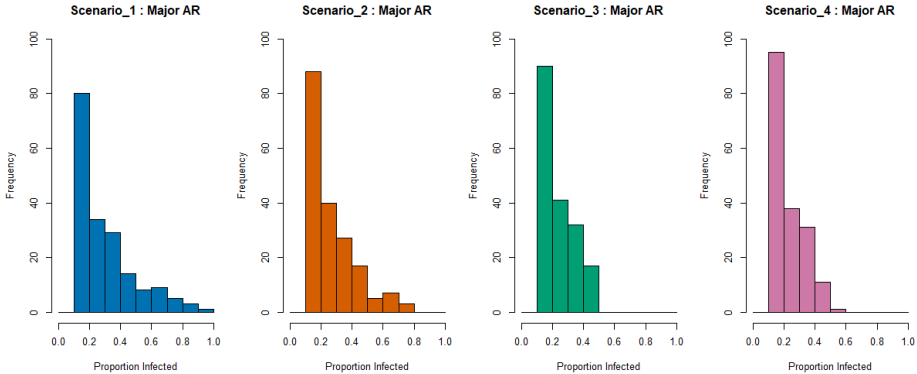


(c) Scenario 3: Strict-cohorting



(d) Scenario 4: Common staff room

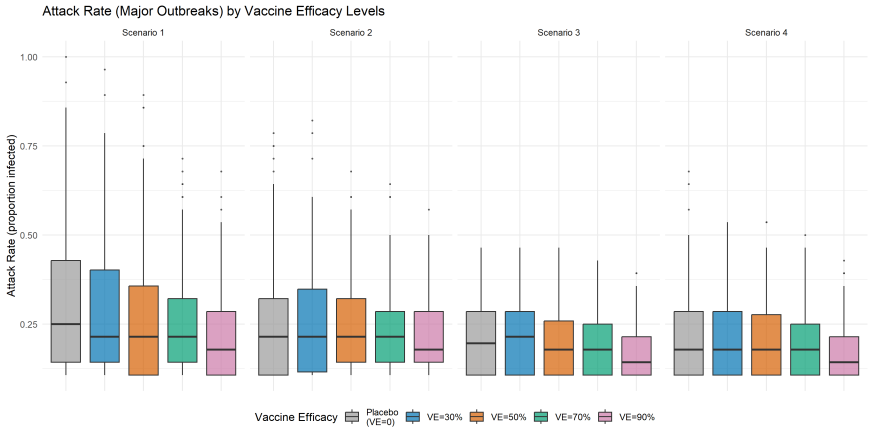
Results: Attack Rate Distributions



Attack rates of major outbreaks in four scenarios

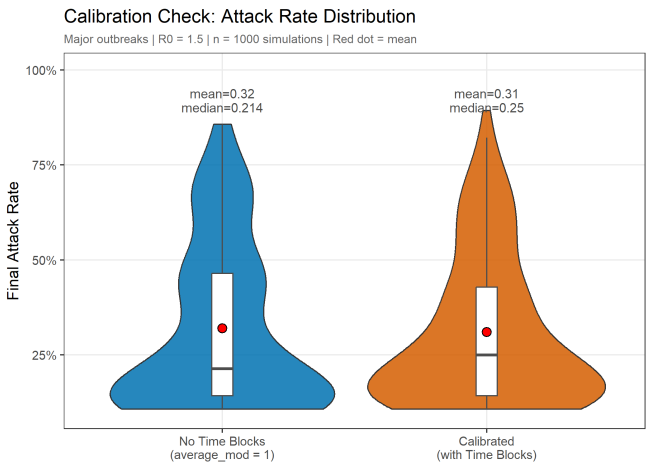
Scenario 1 broadest (60–100%), Scenario 3 most contained (<50%). Mixing structure affects severity, not initiation.

Results: Vaccination Impact



Attack rates of major outbreaks by Vaccine efficacy levels

Calibration Check



Attack rate distributions between with and without time blocks

Key finding

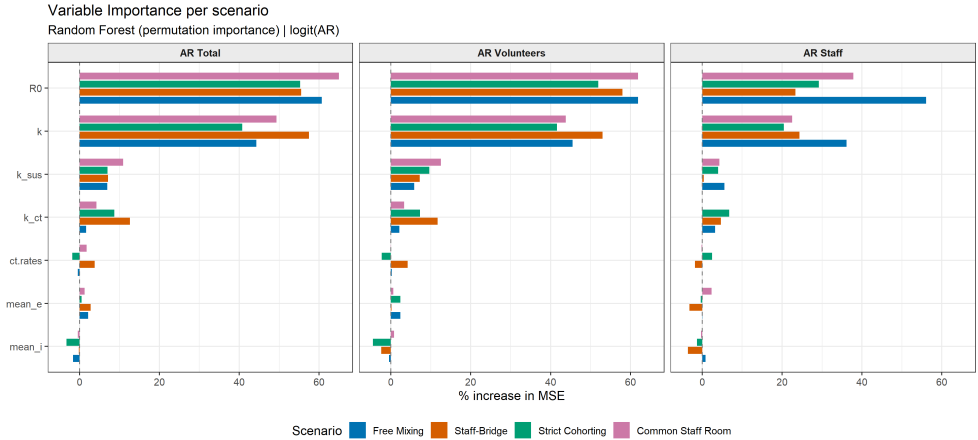
Mean AR: comparable ($\approx 32\%$) in both conditions

Median AR: 21.4% (no blocks) vs. 25.0% (calibrated)

Calibrated model produces fewer extreme outbreaks ($AR > 75\%$) due to $\bar{M}_{total} \approx 1.065 > 1$, which slightly reduces Q.

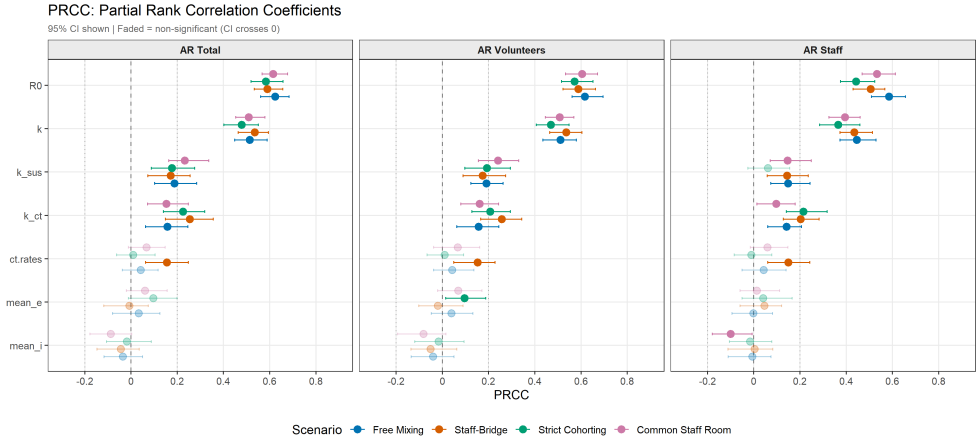
$\Rightarrow$ Time-block calibration preserves R_0 without systematic bias.

SA: Variable Importance



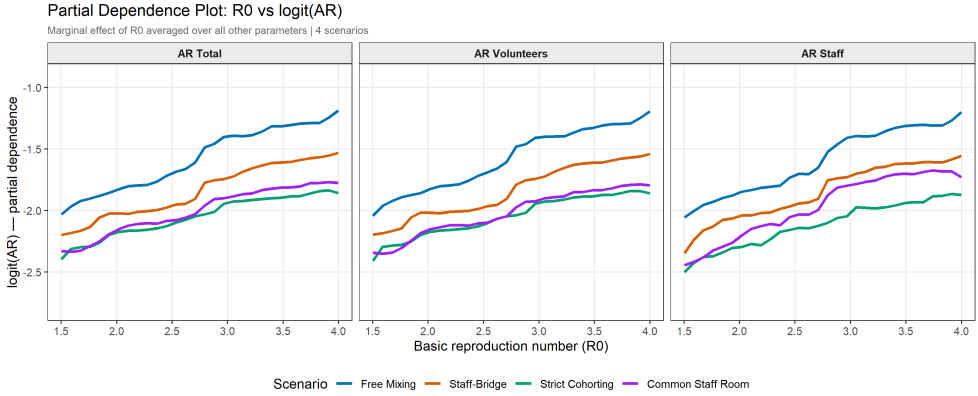
Variable Importance per scenario (%IncMSE)

SA: Partial rank correlation coefficients



Partial Rank Correlation Coefficients per scenario

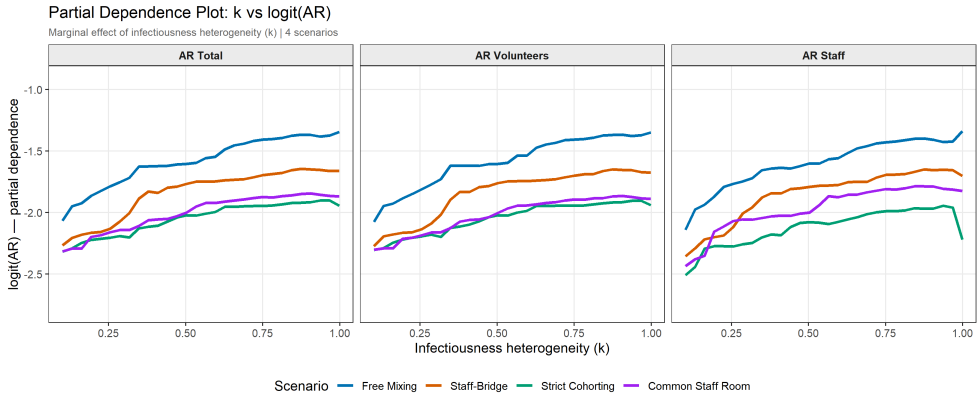
SA: Partial Dependence Plots: R_0



Partial Dependence Plot: logit(AR) vs. R_0

Monotone increasing: steepest at $R_0 \in [1.5, 2.5]$, followed by progressive flattening at higher values.

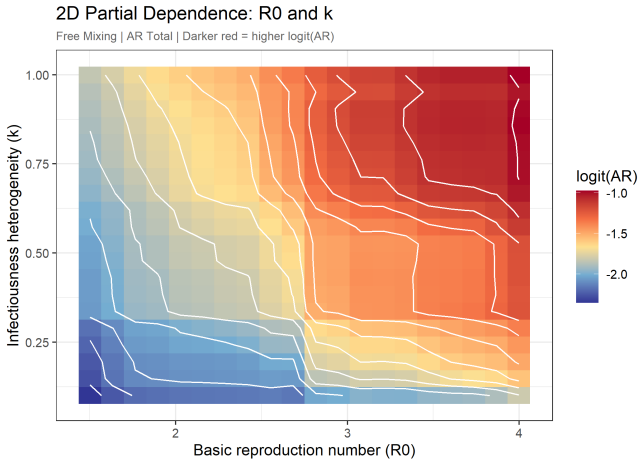
SA: Partial Dependence Plots: k



Partial Dependence Plot: logit(AR) vs. k

Sharp rise at $k \in [0.1, 0.5]$; near-plateau beyond $k = 0.75$. Low $k \Rightarrow$ superspreader dynamics dominate.

SA: Partial Dependence Plots: R_0 and k



Partial Dependence Plot: R_0 and k

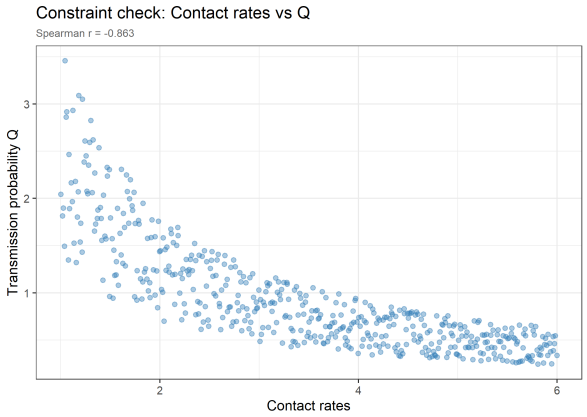
Key finding

Low k (< 0.3): AR remains low even as $R_0 \to 4.0$
$\Rightarrow$ superspreader dynamics govern outcomes

High k (≥ 0.5): AR increases rapidly with R_0
$\Rightarrow$ transmission becomes more predictable

Highest risk: $R_0 > 3$, $k > 0.75$ ($\text{logit}(AR) \approx -1.2$, upper-right quadrant)

SA: Relationship Contact rates vs. Q



Relationship between contact rates and transmission probability Q

Key finding

Spearman $r = -0.863$, $p < 0.001$

$Q = R_0/(\bar{\lambda} \times \bar{M}_{\text{total}})$

Scatter reflects variation in $R_0 \in [1.5, 4.0]$

$\bar{\lambda} \uparrow \Rightarrow Q \downarrow$ to preserve R_0

Effects partially offset $\Rightarrow$ low RF & PRCC importance for *ct.rates*

Not epidemiologically unimportant — effect absorbed by compensating Q

Discussion & Conclusion

Contact structure

Severity of outbreaks is influenced by mixing contact structures. **Scenario 3** Strict-cohorting yields the highest effectiveness in preventing epidemic transmission.

Vaccination

Partial vaccination of volunteers provides further protection, especially useful where cross-contact pathways between groups still exist.

Key drivers

R_0 and k are consistently identified as dominant factors of outbreak dynamics — highlighting the importance of characterising pathogen transmissibility and superspreading potential for risk assessment.

Consistency with literature

Low $k \Rightarrow$ highly variable, extinction-dominated dynamics, consistent with overdispersion patterns reported for SARS-CoV-2 [J. O. Lloyd-Smith et. al. (2005)].

These findings offer a quantitative foundation for guiding staffing plans, vaccination guidelines, and quarantine procedures in the design and conduct of human infection trials at Vaccinopolis.

Limitations

Limitations

- **Small population** ($n = 28$) $\Rightarrow$ high stochastic variability; results not generalisable to larger communities without extensive statistical validation
- **Static parameters**: core epidemiological parameters held constant; does not account for biological variation across viral variants or host immune health
- **Vaccination**: immediate full effect assumed; no time lag for antibody development, no waning immunity, no inter-brand variation ($\varepsilon_{\text{inf}} = 0$)
- **Contact multipliers & mixing matrices**: assumption-based, not empirically measured within Vaccinopolis
- **COVID-like parameterisation**: LHS ranges cover COVID-like and influenza-like profiles, but no explicit pathogen-specific scenario analyses for RSV or influenza

Thank you

Questions & Discussion